



Liquid fuels demand and supply model

Manish handover

6 October focus | Source and reconcile the data

6 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Durban | vopak.com



Source the data before quantifying the fuel levers

Today: integrity, traceability, provincial update and a matched fuel balance

Priority	Manish does	Return for review
1 Data integrity	Investigate incomplete provincial totals, duplicate refinery/history keys and unsupported baselines.	Flag; old value; source page/cell; proposed value; reason; resolved or open.
2 Traceability	Trace priority data from publisher to original file, extract and consuming function.	Exact paths; API/download/manual; units; coverage; used, staged or reporting-only.
3 Provincial update	Verify newer observed provincial petrol/diesel sales, then seek 2025 and latest 2026 coverage.	Province/product/period; observed versus estimated; national tie; missing periods.
4 Production and trade	Reconcile production, imports, exports and stock movements for the same product and period.	Matched litres; source definitions; stock treatment; unexplained residual.
5 Driver evidence	Source agriculture, manufacturing, mining, passenger/freight, power, EV and plant series.	Original evidence and extracts for each chart; revisions and coverage. Scenario settings follow.

Ways of working: Nigel on main; Manish on a named branch from current origin/main. Nigel reviews changes before integration.

Open the exact files and follow their data trail

All paths are relative to your clone of liquid_fuels_model

Start here	Exact path beneath the project folder	What to check
Audit and direction	output/delivered/ Liquid_fuels_source_audit_2026_10_05.xlsx source_profile_2026_10_05.html	Direction, Provincial gaps, Repeated keys; input access and consuming functions.
Provincial demand	assumptions/2026/timeseries/ fuel_sales_department_by_province.csv fuel_sales_department_by_province_quarterly.csv	Incomplete quarters and annual totals; source_file identifies the original workbook.
Production and trade	assumptions/2026/timeseries/ energy_balance_department.csv fuel_trade_fiasa.csv fuel_trade_department_review.csv	Same product/period/units; 2024 diesel imports: 14.793 versus 10.8 bn litres remains a source disagreement.
Duplicate records	assumptions/2026/timeseries/ refinery_production.csv historical_demand.csv	558 extra refinery rows and 15 extra historical rows. refinery_production.csv feeds utilisation, despite its name.
Baseline parameters	assumptions/2026/ vehicles.yaml; agriculture.yaml industrial.yaml; marine.yaml	Source of mileage, fuel mix, baselines and elasticities. Activity indices cannot silently replace parameters.

Raw = original PDF, Excel or API response in external/. Extract = CSV in assumptions/. Calculation = src/lfm/model/.

Run one fetch command for each priority dataset

Use the staging setup on the next page before running these commands

```
$py = '.\.venv\Scripts\python.exe'
```

Dataset	PowerShell command	Main extract
Department	& \$py -m lfm.scripts.fetch_energy_dept --vintage 2026	Sales, provinces, balances, prices
FIASA	& \$py -m lfm.scripts.fetch_fuel_sales --vintage 2026	Sales, imports and exports
Economy	& \$py -m lfm.scripts.fetch_economy --vintage 2026	GDP, population, Treasury; Stats SA workbook on disk
NaTIS	& \$py -m lfm.scripts.fetch_natis --vintage 2026	Fleet stock and registrations
naamsa	& \$py -m lfm.scripts.fetch_naamsa --vintage 2026	BEV, PHEV, hybrid and market sales
Eskom	& \$py -m lfm.scripts.fetch_eskom --vintage 2026	Eskom and IPP OCGT generation

Combined refresh: & \$py -m lfm.scripts.refresh_sources --vintage 2026

Runs the six fetchers plus ACSA. Logs failed commands and compares sales; it does not collect every missing source. Append --offline only to reread originals already on disk. Capture each exit code and warnings.

Keep candidate refreshes separate and test each source

Candidate files first; reviewed promotion into the vintage follows later

PowerShell setup

```
$stage = "runs/manish_fetch_$(Get-Date -Format yyyyMMdd_HH:mm:ss)"  
New-Item -ItemType Directory "$stage/assumptions" -Force | Out-Null  
Copy-Item assumptions/2026 "$stage/assumptions/2026" -Recurse  
$oldA = $env:LFM_ASSUMPTIONS_DIR  
$oldD = $env:LFM_DATA_DIR  
$env:LFM_ASSUMPTIONS_DIR = (Resolve-Path "$stage/assumptions").Path  
$env:LFM_DATA_DIR = Join-Path (Get-Location) "$stage/data"  
New-Item -ItemType Directory "$stage/data/raw" -Force | Out-Null
```

Stats SA manual prerequisite

Download the GDP Time series workbook into \$stage/data/raw/statssa/ before the economy command. The reader uses the Annual sheet; quarterly 2026 extraction needs support. Existing raw caches may be copied into the candidate raw folder.

Checks and restoration

Compare old/new values, dates, units, duplicates and completeness. Preserve prior data on failed or partial fetches. Add tests for new parser support.

Restore in finally if scripted:

```
$env:LFM_ASSUMPTIONS_DIR = $oldA  
$env:LFM_DATA_DIR = $oldD
```

Then run:

```
& $py -m lfm check --vintage 2026  
& $py -m pytest -q
```

A successful command is not numerical verification or adoption.

Close the retrieval gaps and return reviewable evidence

The LLM implements and tests missing extraction; Manish reviews source fidelity

Missing coverage	Source and implementation needed	Return today
Provincial 2023 onward	Department market overview has charts through 2024. Verify PDF/table values; seek 2025 and 2026 YTD. Current workbook fetcher does not parse these charts.	Observed/estimated province-product table and national tie; explicit gaps.
Stats SA driver series	P0441 GDP/agriculture; P3041.2 manufacturing; P2041 mining; P7162 land transport. Add current downloads/parsers; activity is not fuel litres.	Originals, extracts, dates, units, geography, source table and revision flags.
Detailed fuel trade	SARS petrol/diesel tariff lines with quantities/units; energy/operator production and stock records. FIASA fetch does not download SARS.	Production + imports - exports - stock build; consumption comparison and residual.
Data sharing and review	247 prior downloads and the stand-up transcript remain local after upload review blocks. Shared CSVs do not prove originals are available or verified.	Missing-original list; per-source outcomes; code and tests on Manish branch.

Afternoon handback: source map, resolved/open flags, provincial coverage, matched balance and driver data. Nigel updates the SCR pack; lever scenarios follow sourcing.

[Open detailed instructions and source links](#)

Return sourced data and explicit gaps

Review evidence before setting lever
scenarios

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